

SAHIL RAOSAHEB SONAWANE

Computer Engineering Student | AI/ML • GenAI • Backend Development

[Email](#) | [GitHub](#) | [LinkedIn](#) | [Portfolio](#)

SUMMARY

Third-year Computer Engineering student building at the intersection of AI/GenAI and software engineering. My work spans LLM applications, backend systems, GIS, and computer vision, with projects taken from ideas to working prototypes and hackathon presentations.

EDUCATION

KJEI's Trinity College of Engineering and Research, Pune

B.E. Computer Engineering, Third Year (T.E.)

Expected graduation: 2028

FY SGPA: 7.86 | SY SGPA: 7.94

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript

AI / GenAI: Machine Learning, LLM applications, Gemini API

Backend / APIs: FastAPI, Node.js, Express.js, REST APIs

Frontend: React, Vite, Three.js

Data / Databases: SQLite, Pandas, NumPy

Computer Vision / GIS: OpenCV, MediaPipe, Shapely

Testing / Tools: Pytest, Pydantic, Git, GitHub

RESEARCH

Agentic Learning Digital Twin: A Framework for Intelligent Personalized Learning

[\[Paper\]](#)

Published Research Paper | International Journal of Engineering Development and Research (IJEDR) | Vol. 14, Issue 3 | Sep. 2026 | pp. 652–660
| Paper ID: IJEDR2603606

Authors: Sahil Raosaheb Sonawane, Prof. Santosh S. Shinde, Prof. Madhavi Netke

- Proposed a six-layer Agentic Learning Digital Twin framework connecting persistent learning context, learner-state representation, agentic reasoning, personalized intervention, knowledge access and governance.
- Built MnemOS as a practical prototype of selected ALDT components: learning-memory capture, persistent storage, contextual retrieval, timeline access and Gemini-based assistance.

TECHNICAL PROJECTS

MnemOS | AI learning-memory system (working prototype)

[GitHub repo](#)

- Built a Manifest V3 Chrome extension that captures study activity and stores it in SQLite through a Node.js/Express API, with a React 19, TypeScript and Vite dashboard (learning timeline, Recharts charts). Frontend runs on Vercel, backend on Render.
- Added a Gemini 2.5 Flash assistant that answers questions like "What have I been learning recently?" using the stored history as context, and generates skill trends and career suggestions.

SURAKSHA-SETU | Flood and landslide decision-support prototype (SIH26191)

[GitHub repo](#)

- Built a FastAPI backend with 11 REST endpoints and SQLite. Hazard and habitation data goes through GeoJSON analysis in Shapely (point-in-polygon, intersections, Haversine distance) and a deterministic risk engine that outputs a 0–100 score, risk level and mitigation priority.
- Wrote a five-stage relocation engine that screens shelters, validates capacity against the affected population and recommends evacuation. Includes Pydantic validation, Pandas/NumPy ingestion, Pytest tests and a React frontend; runs on synthetic demo data.

ECO-ROUTE | Plastic-leakage decision-support platform for Mumbai | Project in Progress

- Developing a workflow from citizen report to response tracking (FastAPI, SQLite, React/Vite, role-based Citizen and Government access) in which ML-based plastic detection and classification with confidence estimates verifies each report.
- Using Leaflet, OpenStreetMap, the Overpass API and Shapely to identify potential plastic escape pathways from mapped drainage and waterway connectivity, with density-based clustering on density, recency and growth to detect hotspots.
- Implementing Priority-v2, a weighted score that combines hotspot, GIS and rainfall/weather inputs to rank urgency, plus constraint-based optimization to support intervention choice under resource limits.

GestureVerse | Gesture-controlled particle worlds

[GitHub repo](#)

- MediaPipe hand tracking switches between six particle scenes from webcam gestures; about 38,000 particles rendered in Three.js with custom WebGL shaders and bloom.

HACKATHONS & LEADERSHIP

Team Leader across all listed competitions

- Smart India Hackathon:** participated twice; selected for the second round in one of them.
- FarAway Hackathon:** selected for the pre-final rounds. **Aavishkar** (university-level): selected for the second round.
- InnovateX, IIT Bombay Techfest 2026 (Mumbai Zonal):** presented the project at IIT Bombay. **Codemble:** participated.